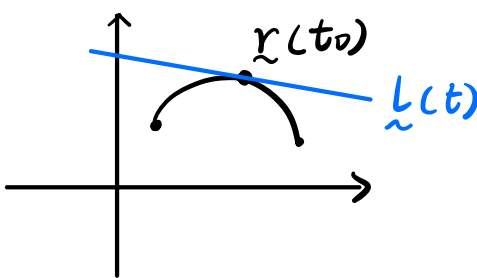


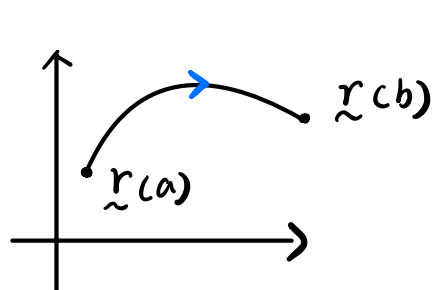
Week 1

- $\underline{r}(t) = \begin{bmatrix} r_1(t) \\ r_2(t) \\ r_3(t) \end{bmatrix}$

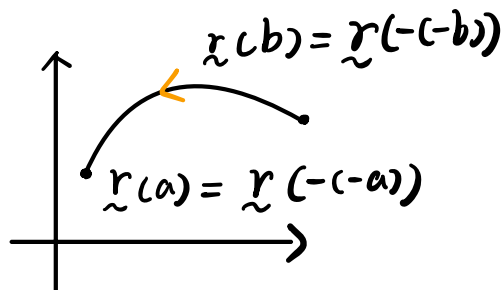
$$\underline{v}(t) = \frac{d}{dt} \underline{r}(t), \quad \underline{a}(t) = \frac{d^2}{dt^2} \underline{r}(t)$$

-  $\Rightarrow \underline{l}(t) = \underline{r}(t_0) + t \cdot \underline{r}'(t_0)$
Physics, $S = S_0 + vt$

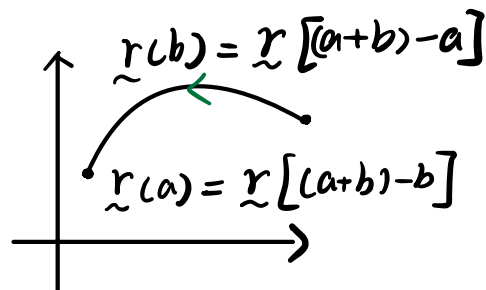
- $$\begin{cases} \underline{r}(t), t \in [a, b] & \textcircled{1} \\ \underline{r}(-t), t \in [-b, -a] & \textcircled{2} \\ \underline{r}[(a+b)-t], t \in [a, b] & \textcircled{3} \end{cases}$$



move from $\underline{r}(a)$ to $\underline{r}(b)$
when t from a to b



move from $\underline{r}(b)$ to $\underline{r}(a)$ when t
from $-b$ to $-a$

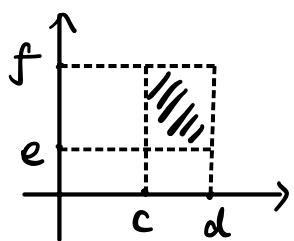


move from $\underline{r}(b)$ to $\underline{r}(a)$ when t
from a to b

- $$\begin{cases} \text{dot product} = \underline{a} \cdot \underline{b} = \|\underline{a}\| \|\underline{b}\| \cos \theta \\ \text{cross product} = \|\underline{a} \times \underline{b}\| = \|\underline{a}\| \|\underline{b}\| \sin \theta \end{cases}$$

- $F_{yx}(a,b) = F_{xy}(a,b)$ if F_{yx} and F_{xy} is continuous (a,b)

Proof: Integral $F_{yx}(a,b)$ at neighborhood



$$\iint_{[c,d] \times [e,f]} F_{xy}(x,y) dy dx$$

$$= \int_c^d \int_e^f \frac{\partial}{\partial y}(F_x) dy dx$$

$$= \int_c^d F_x(x,y) \Big|_e^f dx$$

$$= \int_c^d [F_x(x,f) - F_x(x,e)] dx$$

$$= [F(x,f) - F(x,e)]_c^d$$

$$= F(d,f) - F(d,e) - F(c,f) + F(c,e)$$

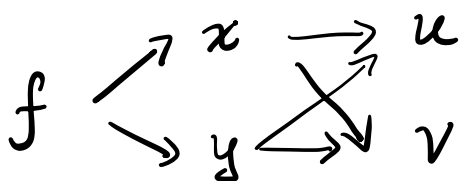
同理 F_{xy} 和 F_{yx} 一样

- chain rule. $f(x,y)$, $x(u,v)$, $y(u,v)$

$$\therefore \frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\begin{cases} \frac{\partial f}{\partial v} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial v} \end{cases}$$

$$\therefore \begin{bmatrix} \frac{\partial}{\partial u} \\ \frac{\partial}{\partial v} \end{bmatrix} f = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} \cdot f$$



- directional derivative

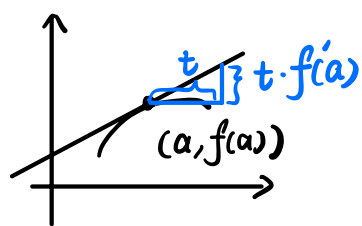
$$\underline{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \quad \nabla f(x, y, z) = \begin{bmatrix} \partial f / \partial x \\ \partial f / \partial y \\ \partial f / \partial z \end{bmatrix}$$

Application: $f(x_0, y_0, z_0) + t \underline{v}$, then

$$\begin{aligned} \frac{d}{dt} f|_{t=0} &= \frac{\partial f}{\partial x} \cdot \frac{d(x_0 + at)}{dt} + \dots \\ &= \frac{\partial f}{\partial x} \cdot a + \frac{\partial f}{\partial y} \cdot b + \frac{\partial f}{\partial z} \cdot c \\ &= \underline{v} \cdot \nabla f = D_{\underline{v}} f \end{aligned}$$

- Equation of line / plane

① line:



$$\underline{r}(t) = \underline{r}(a) + t \underline{r}'(a)$$

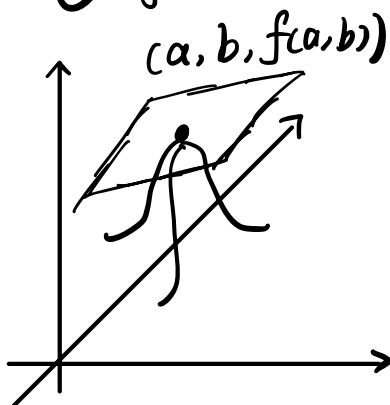
$$\therefore \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} a \\ f(a) \end{bmatrix} + t \begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix}$$

$$t = x - a, \quad \text{or } \lambda \text{ (2)}$$

$$= \begin{bmatrix} a \\ f(a) \end{bmatrix} + t \begin{bmatrix} 1 \\ f'(a) \end{bmatrix}$$

$$\therefore \begin{cases} x = a + t \\ y = f(a) + t f'(a) \end{cases} \quad \text{①} \Rightarrow y = f(a) + (x - a) f'(a) \quad \text{②}$$

② plane:



$$\underline{r}(\alpha, \beta) = \underline{r}(a, b) + \alpha \frac{\partial \underline{r}}{\partial x}(a, b) + \beta \frac{\partial \underline{r}}{\partial y}(a, b)$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ b \\ f(a, b) \end{bmatrix} + \alpha \begin{bmatrix} 1 \\ 0 \\ f_x \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ f_y \end{bmatrix}$$

$$\therefore \begin{cases} x = a + \alpha \\ y = b + \beta \\ z = f(a, b) + \alpha f_x + \beta f_y \end{cases} \Rightarrow \begin{aligned} &z - f(a, b) \\ &= f_x(x - a) + f_y(y - b) \end{aligned}$$

Week 2

- Let $F(x, y, z) = f(x, y) - z = 0$

then $\nabla F = \begin{bmatrix} f_x \\ f_y \\ -1 \end{bmatrix} \perp$ level surface $F(x, y, z) = 0$

- A motion through (a, b, c) , $\underline{x}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$

and $\underline{x}(0) = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \Rightarrow \underline{v}(t) = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}$ 面上曲线的变化方向与 ∇F 垂直

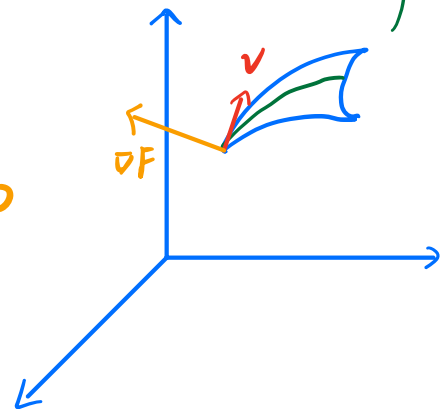
Let: $F(x, y, z) = 0 \rightarrow$ level surface

On the level surface, $F(x, y, z) = 0$ doesn't change,

means $\frac{\partial F}{\partial t} = 0$

$$\Rightarrow \frac{\partial F}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial F}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial F}{\partial z} \cdot \frac{dz}{dt} = 0$$

$$\therefore \nabla F \cdot \underline{v}(t) = 0$$



- $f(x, y) = f(a, b) + \begin{bmatrix} f_x \\ f_y \end{bmatrix} \begin{bmatrix} x-a \\ y-b \end{bmatrix} + \begin{bmatrix} x-a & y-b \end{bmatrix} \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \begin{bmatrix} x-a \\ y-b \end{bmatrix} + \dots$

For $\begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = A \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \cdot A^T$ where $A \cdot A^T = I$.

Here: $\begin{cases} \text{local max: } \lambda_1 \leq \lambda_2 < 0 \\ \text{local min: } \lambda_1 \geq \lambda_2 > 0 \\ \text{saddle: } \lambda_1 < 0 < \lambda_2 \\ \text{inconclusive: } \lambda_1 = \lambda_2 = 0 \end{cases}$

• Lagrange Multiplier Method

Condition: $F(x, y, z) = 0$, find extrema $u = f(x, y, z)$.

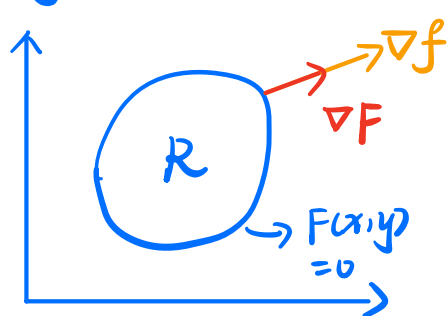
Then solve
$$\begin{cases} \nabla [f(x, y, z) - \lambda F(x, y, z)] = 0 & \textcircled{1} \\ F(x, y, z) = 0 & \textcircled{2} \end{cases}$$

From ①: $\nabla f = \lambda \nabla F$ $D_{\underline{v}} f = \underline{v} \cdot \nabla f$. When $\underline{v}$ tangent to $\{F=0\}$, $D_{\underline{v}} f = \underline{v} \cdot \nabla f = \underline{v} \cdot \lambda \nabla F = 0$

$\therefore \nabla F \perp \{F=0\}$
 $\therefore \nabla f \perp \{F=0\}$

→ 变化最快的方向与限制条件的法向量共线

Specially, in 3-D space:



region $R: F(x, y) = 0$

$z = f(x, y)$ extrema on R .

Then: $\nabla f = \lambda \nabla F$

e.g. $h(x, y) = x^2 + y^2 - x^4$, $x \in [-2, 2]$, $y \in [-2, 2]$

Find critical points.

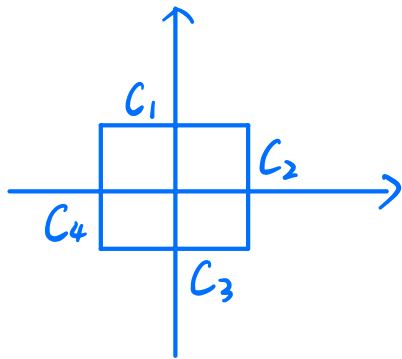
Solution:
$$\begin{cases} h_x = 2x - 4x^3 = 0 \\ h_y = 2y = 0 \end{cases} \Rightarrow (0, 0), (\frac{\sqrt{2}}{2}, 0), (-\frac{\sqrt{2}}{2}, 0)$$

$$\Rightarrow \begin{cases} h_{xx} = 2 - 12x^2 \\ h_{xy} = 0 \\ h_{yy} = 2 \end{cases} \Rightarrow \begin{bmatrix} h_{xx} & h_{xy} \\ h_{yx} & h_{yy} \end{bmatrix} = \begin{bmatrix} 2 - 12x^2 & 0 \\ 0 & 2 \end{bmatrix}$$

① $x=0, y=0, \lambda_1 > \lambda_2 > 0 \Rightarrow \text{local min}$

② $x = \frac{\sqrt{2}}{2} \text{ or } -\frac{\sqrt{2}}{2}, y=0, \lambda_1 < 0 < \lambda_2 \Rightarrow \text{saddle point}$

Then consider the boundary $\rightarrow x^2 \in [0, 4]$



$$C_1 = (x, 2), x \in [-2, 2]$$

$$\begin{aligned} \therefore h(x, y) &= x^2 + y^2 - x^4 \\ &= x^2 - x^4 + 4 \\ &= x^2(1 - x^2) + 4 \in [-8, 4\frac{1}{4}] \end{aligned}$$

Similarly, consider C_2, C_3, C_4 in the same way

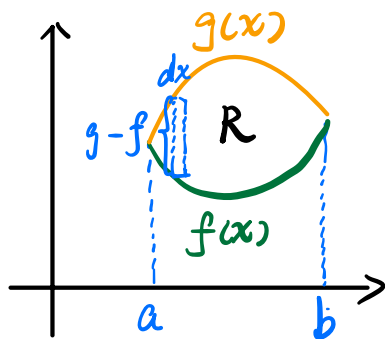
• Jacobian Theorem

$$\left\{ \begin{aligned} \frac{\partial(x, y)}{\partial(u, v)} &= \begin{bmatrix} \partial x / \partial u & \partial x / \partial v \\ \partial y / \partial u & \partial y / \partial v \end{bmatrix} \\ \frac{\partial(u, v)}{\partial(x, y)} &= \begin{bmatrix} \partial u / \partial x & \partial u / \partial y \\ \partial v / \partial x & \partial v / \partial y \end{bmatrix} \end{aligned} \right.$$

Week 3

- Multiple Integral: $\iint f(t) dt dt = f(t) + C_1 t + C_2$ n次积分出现n个未知数

Cutting and Integral

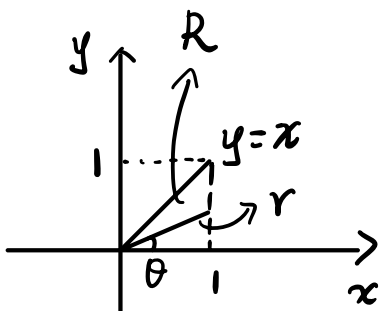


$$R = \int_a^b [g(x) - f(x)] dx.$$

$$x \in [a, b]$$

$$y \in [g(x) - f(x)]$$

e.g.



i) x-first Method.

$$\int_0^1 y dx = \int_0^1 x dx = \frac{1}{2}$$

ii) y-first Method

$$\int_0^1 (1-x) dy = \int_0^1 (1-y) dy = \frac{1}{2}$$

$$iii) r = \frac{1}{\cos \theta} = \sec \theta$$

$$\int_0^{\frac{\pi}{4}} \int_0^{\sec \theta} r dr d\theta = \frac{1}{2} \int_0^{\frac{\pi}{4}} \sec^2 \theta d\theta = \frac{1}{2}$$

Riemann Sum:

其中假设 Z_k 为最小分割法.

$$\text{例: } \left\{ \begin{aligned} \sum_{i=1}^n f(\tilde{x}_i) (x_i - x_{i-1}) &= \sum_{k=1}^N f(\tilde{x}_i) (Z_k - Z_{k-1}) \quad \text{①} \\ \sum_{j=1}^m f(\tilde{w}_j) (w_j - w_{j-1}) &= \sum_{k=1}^N f(\tilde{w}_j) (Z_k - Z_{k-1}) \quad \text{②} \end{aligned} \right.$$

$\tilde{x}_i$ 和 x_{i+1} 中任意一点



①: divide according to x_i

②: divide according to w_i .

①-②: $\int$ LHS = difference between dividing by x and w

$$\text{RHS} = \sum_{k=1}^n [f(\tilde{x}_i) - f(\tilde{w}_j)] (z_k - z_{k-1})$$

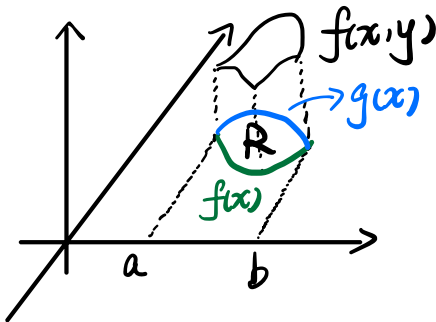
Suppose it's continuous:

$$|\tilde{x}_i - \tilde{w}_j| \leq \delta \Rightarrow |f(\tilde{x}_i) - f(\tilde{w}_j)| \leq \epsilon$$

$$\therefore \text{RHS} \leq \epsilon \sum_{k=1}^n (z_k - z_{k-1}) = \epsilon (b - a)$$

$\Rightarrow$ no matter how we divide, we can make their differences as small as possible

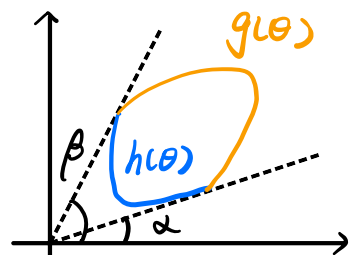
• Double integral.



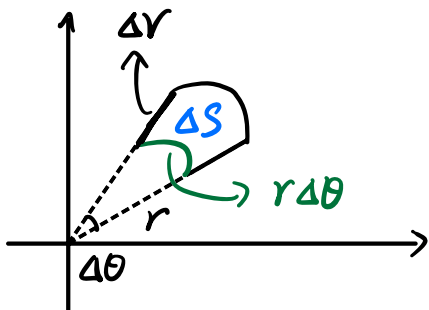
$$V = \iint f(x,y) dA$$

$$= \int_a^b \int_{f(x)}^{g(x)} f(x,y) dy dx$$

($\rightarrow$ 先沿 x 轴 后沿 y 轴)



• Polar form of double integral



$$\therefore dS = r dr d\theta$$

$$\therefore \iint_R F(r,\theta) dS = \iint_R F(r,\theta) \cdot r \cdot dr \cdot d\theta$$

$$= \int_{\alpha}^{\beta} \int_{g(\theta)}^{h(\theta)} F(r,\theta) r dr d\theta$$

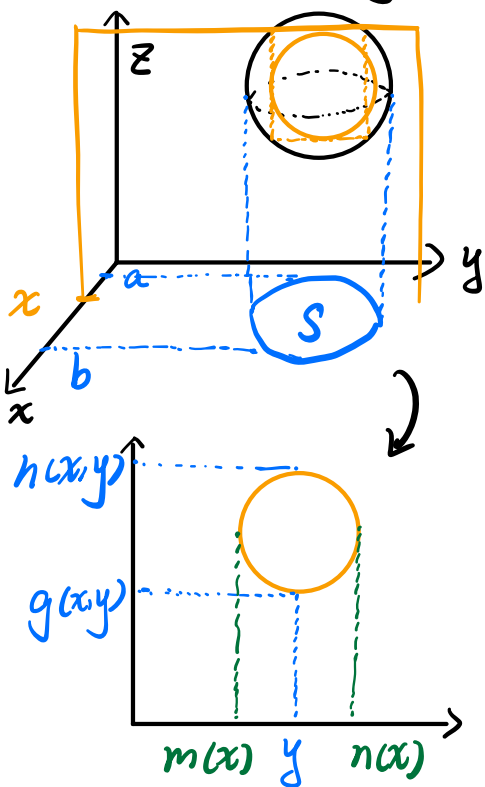
e.g. $I = \int_{-\infty}^{+\infty} e^{-x^2} dx \Rightarrow I^2 = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-(x^2+y^2)} dy dx$

$$\therefore I^2 = \int_0^{2\pi} \int_0^{+\infty} e^{-r^2} r dr d\theta$$

$$= \int_0^{2\pi} d\theta \cdot \int_0^{+\infty} e^{-r^2} d\frac{1}{2}r^2 = \pi$$

$$\therefore I = \int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}$$

• Triple integral



$$\iiint_V F(x,y,z) dx dy dz$$

$$= \int_a^b \left[\iint_S F(x,y,z) dy dz \right] dx$$

$$= \int_a^b \left[\int_{m(x)}^{n(x)} \left(\int F(x,y,z) dz \right) dy \right] dx$$

$$= \int_a^b \int_{m(x)}^{n(x)} \int_{g(x,y)}^{h(x,y)} F(x,y,z) dz dy dx$$

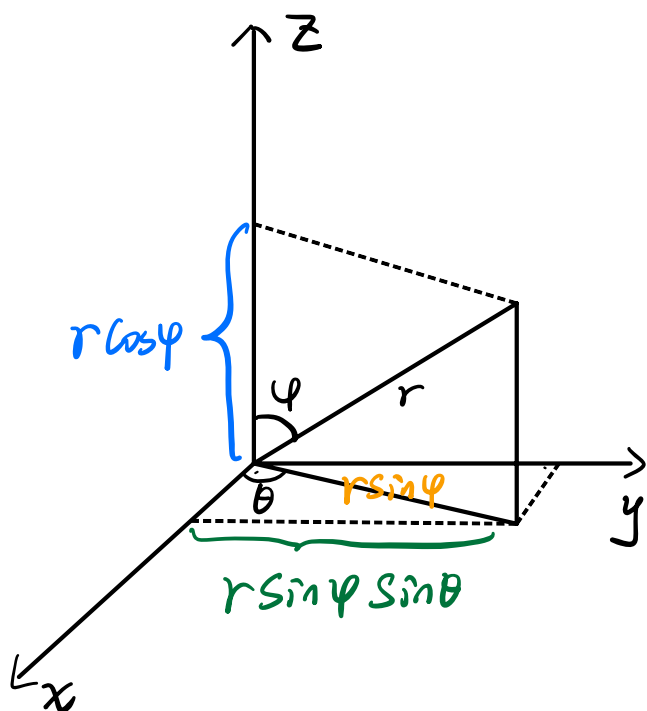
先将三重化成二重再积分

• Triple integral in x-y polar form

$$\iiint_V F dx dy dz = \iiint_V F(r,\theta,z) r dr d\theta dz$$

• Spherical integral

$$\begin{cases} x = r \sin \psi \cos \theta \\ y = r \sin \psi \sin \theta \\ z = r \cos \psi \end{cases}$$



$$\iiint_V F(\varphi, \theta, r) dV$$

z 方向 $x-y$ 面上

$$\Delta V = \Delta r \cdot \boxed{r \cdot \Delta \varphi} \cdot \boxed{r \sin \varphi \cdot \Delta \theta}$$

$$= r^2 \sin \varphi \Delta r \Delta \varphi \Delta \theta$$

$$\therefore \iiint_V F(\varphi, \theta, r) dV$$

$$= \iiint_V F(\varphi, \theta, r) r^2 \sin \varphi dr d\varphi d\theta$$